



Chapter 6 Data Preparation - Data Cleaning, Wrangling and Aggregation

Data Analytics Using Python

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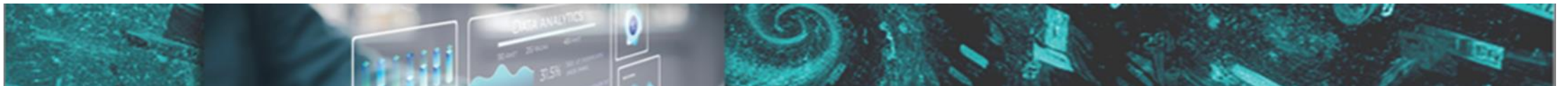


Chapter 6 Data Preparation - Data Cleaning, Wrangling and Aggregation

<https://he.kendallhunt.com/product/data-analytics-using-python>

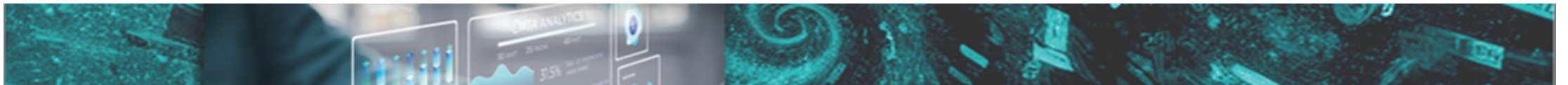
Introduction to Data Preparation

- raw datasets: maybe large, unstructured, and may contain errors, inconsistencies, and missing values
- preprocess the raw data with data cleaning, data wrangling and data aggregation
 - data cleaning: provide solutions for missing values, duplicate values, inconsistent formatting, outliers, errors and inaccuracy and incompleteness
 - data wrangling: transform the raw data into a more structured format for analysis
 - data aggregation: categorize and summarize data using groups with specified criteria



Data Cleaning

- review the raw dataset, identify missing values, errors and inconsistencies, and then correct them so that they are ready for further processing
- tasks that can be done with data cleaning:
 - Identifying and handling missing values
 - Converting data types or formats into consistent types or formats
 - Removing duplicate or irrelevant values
 - Detecting outliers



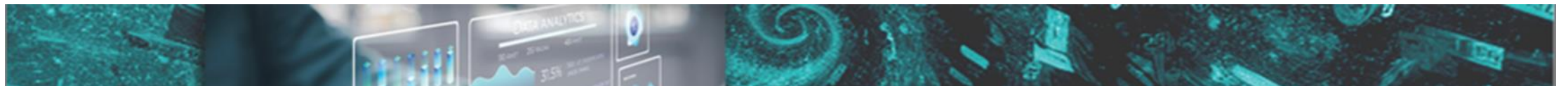
Data Cleaning

- missing data

Data Types	Missing Data Representation
numerical	NaN (Not a Number)
non-numerical	None

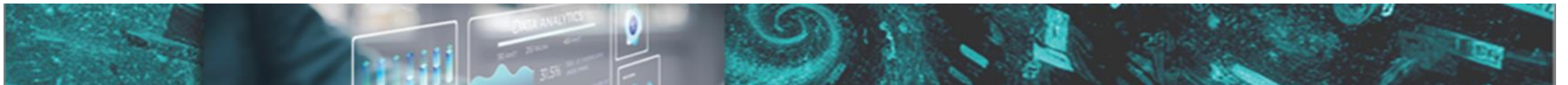
- identify and handle missing data:

Functions	Descriptions related with data cleaning
df.info()	provides the number of non-null values in each column
isnull()	returns True if there is missing value(s) depending on the argument
notnull()	returns True if there isn't missing value(s) depending on the argument
dropna()	returns only the non-null data and index values
fillna()	fill the missing value(s) with specified value(s) such as mean value



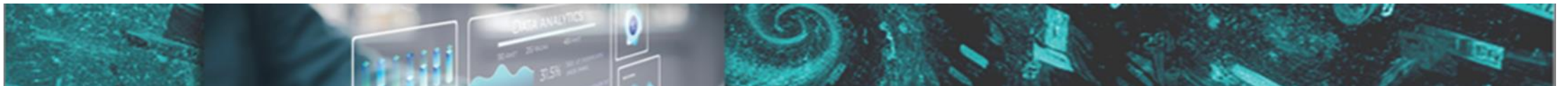
Data Cleaning

- correct inconsistencies in formatting dates:
 - use `pd.to_datetime()` to convert strings to datetime format
 - call `dt.strftime()` (e.g., `'%Y-%m-%d'`) to format date
- identify inconsistencies (commas vs. decimals, leading/trailing spaces) in numbers
 - use string manipulation methods or regular expressions such as `str.replace(',', '')`
 - call `pd.to_numeric()` to convert strings to numeric data types
- remove extra spaces in texts, call `str.strip()`
- other text standardization, use string manipulation methods or regular expressions
- outliers
 - detect outliers, call Dataframe's `describe()` method and `quantile()` method
 - remove outliers from the dataset, call Numpy's `clip()` method to cap them at specified percentiles



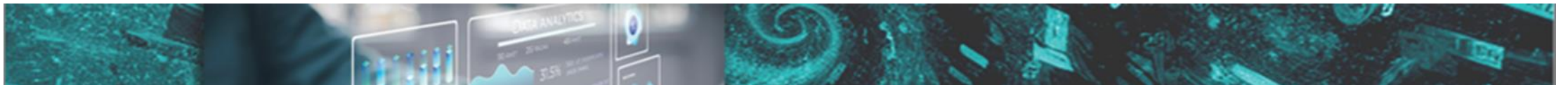
Data Wrangling

- involves data transformation (adding new features, reshaping data, feature scaling), combining data from various sources, and encoding categorical data for analysis
 - transform categorical data into numerical: using techniques like indicator variables or one-hot encoding
 - One-hot encoding creates new binary variables (0 or 1) for each category, except for a chosen reference category. These new variables indicate whether a data point belongs to a specific category
 - One often used function for one-hot encoding is `get_dummies()`.



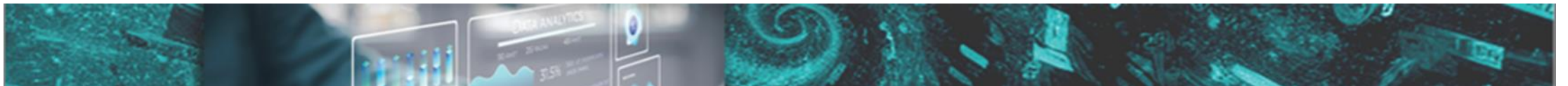
Data Wrangling

- involves data transformation (adding new features, reshaping data, feature scaling), combining data from various sources, and encoding categorical data for analysis
 - Chapter 4 introduced data reshaping
 - pivoting/melting tables
 - stacking/unstacking
 - data reshaping often involves working with hierarchical indexes



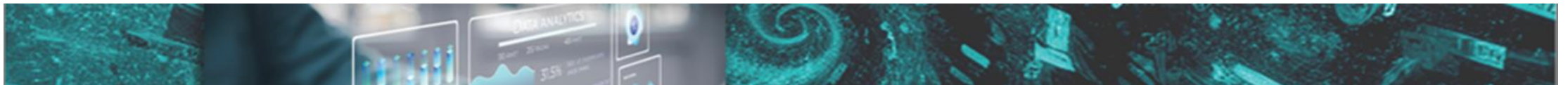
Data Aggregation

- summarize large datasets and identify trends
- involves grouping data points with shared characteristics and applying functions like sum, mean, or count to calculate overall statistics
- simplifies complex data distributions, making trends and patterns easier to identify through visualization analysis



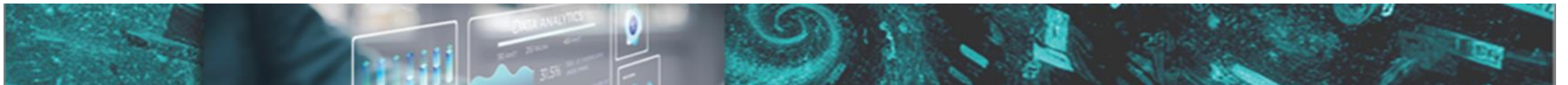
Data Aggregation

functions	descriptions
Descriptive summary functions	
describe()	provide overview of various descriptive statistics for numerical columns in DataFrame; covered in chapter 3(3.3)
value_counts()	get counts for each category for categorical data; covered in chapter 3(3.5 last example)
mean() median() std() min() max() var()	calculate the average (mean), the median (middle value), the standard deviation, the minimum value, the maximum value, the variance of a numerical column; covered in chapter 3 various sections



Data Aggregation

functions	descriptions
Descriptive summary functions	
GroupBy operations	
groupby()	group data by categorical columns and then calculate summary statistics for each group; covered in chapter 4(4.1)
agg()	apply a function or set of functions to each group, and return a new DataFrame
transform()	apply a function to each group, and create a new column for each applied function on the original DataFrame



Data Aggregation

functions	descriptions
Descriptive summary functions	
Build hierarchies for multi-level analysis	
<code>df.set_index()</code>	provide hierarchical Indexing (MultiIndex); covered in chapter 4(4.2)
<code>df.reset_index()</code>	
<code>multiIndex.from_tuples()</code>	
<code>pivot_table()</code>	offer pivot tables; covered in chapter 4(4.3)
<code>crosstab()</code>	
<code>stack()</code>	reshape data; covered in chapter 4(4.3)
<code>unstack()</code>	
<code>melt()</code>	

